



HUST

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HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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The background of the slide is a dark blue field filled with a pattern of red dots. These dots are arranged in a way that they form a large, stylized 'S' shape on the left and a large, stylized 'C' shape on the right, which together enclose the central text. The dots are of varying sizes and are distributed to create a sense of depth and movement.

SOICT

School of Information and Communication Technology

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Natural Language Processing

4 – Large Language Model and Prompt Engineering

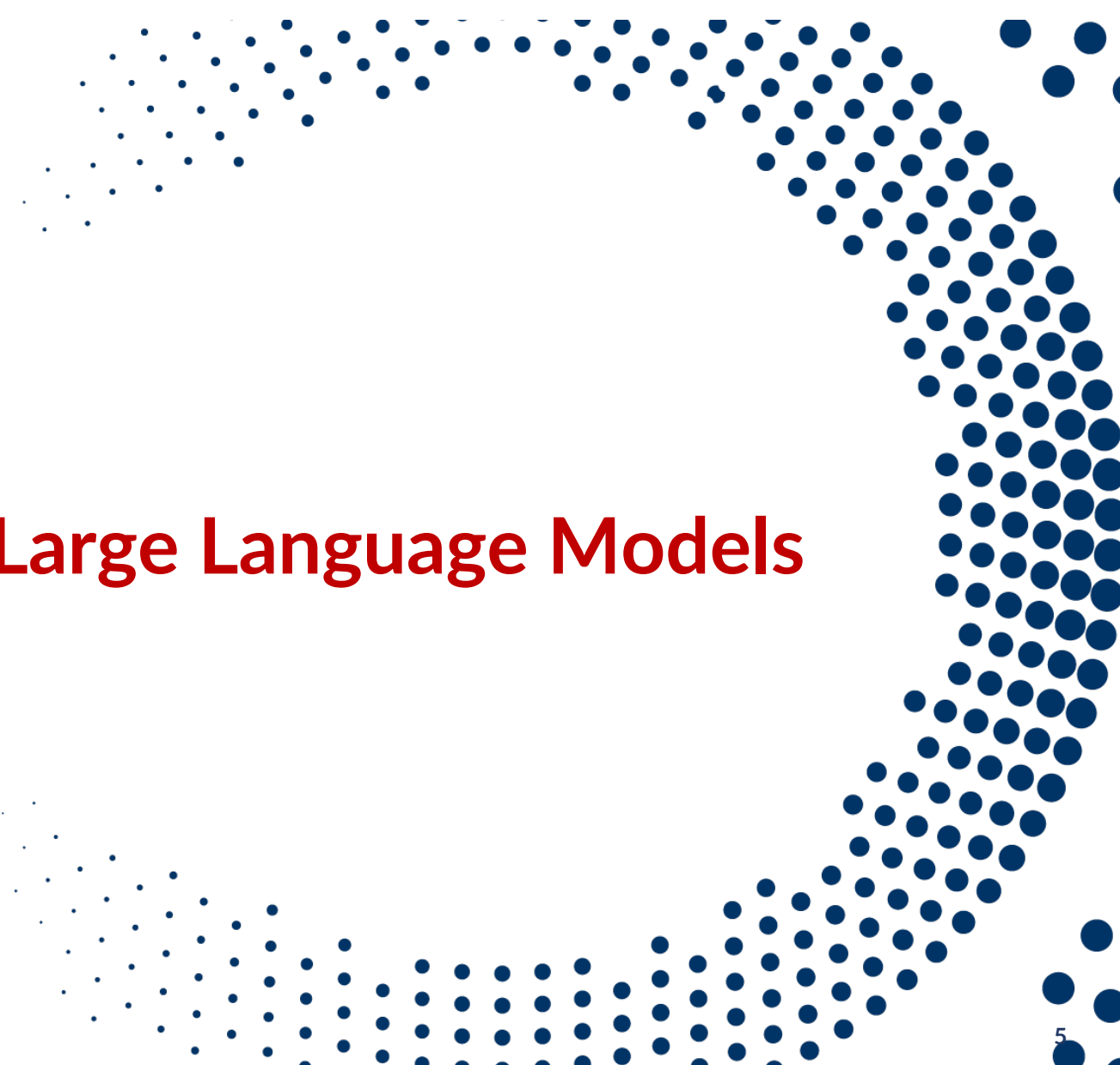
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FYI

- *Lecturer: Dr. Bui Thi Mai Anh*
- **Research domains:**
 - **Software Engineering**
 - **Applied AI in Software Engineering**
 - **Search-based Software Engineering:**
evolutionary computing,
reinforcement learning, deep learning
 - **NLP4SE**
- **Lab: Intelligent Software Engineering – BK-AI Center**

<https://iselab-dorlab.github.io/>

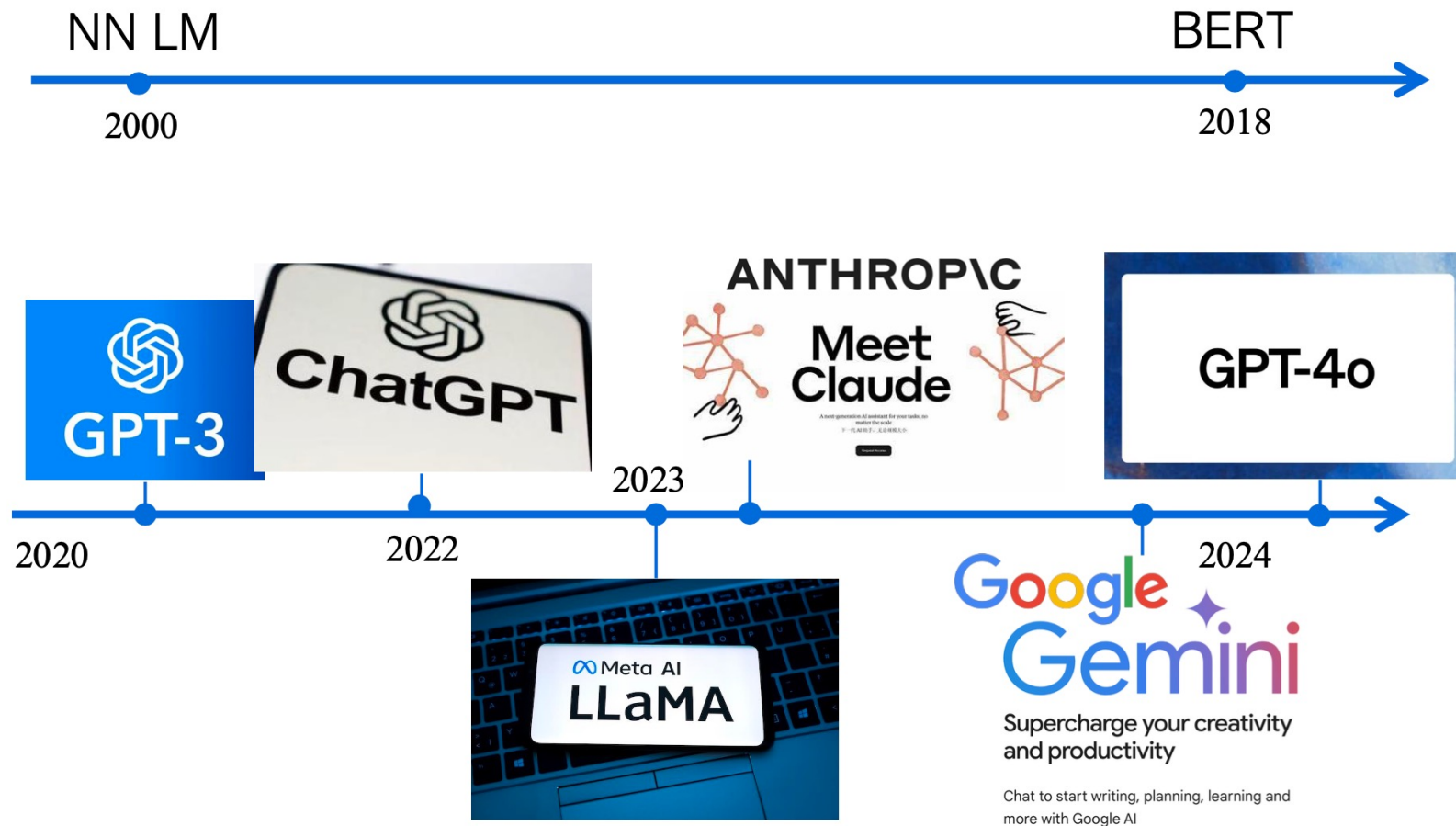




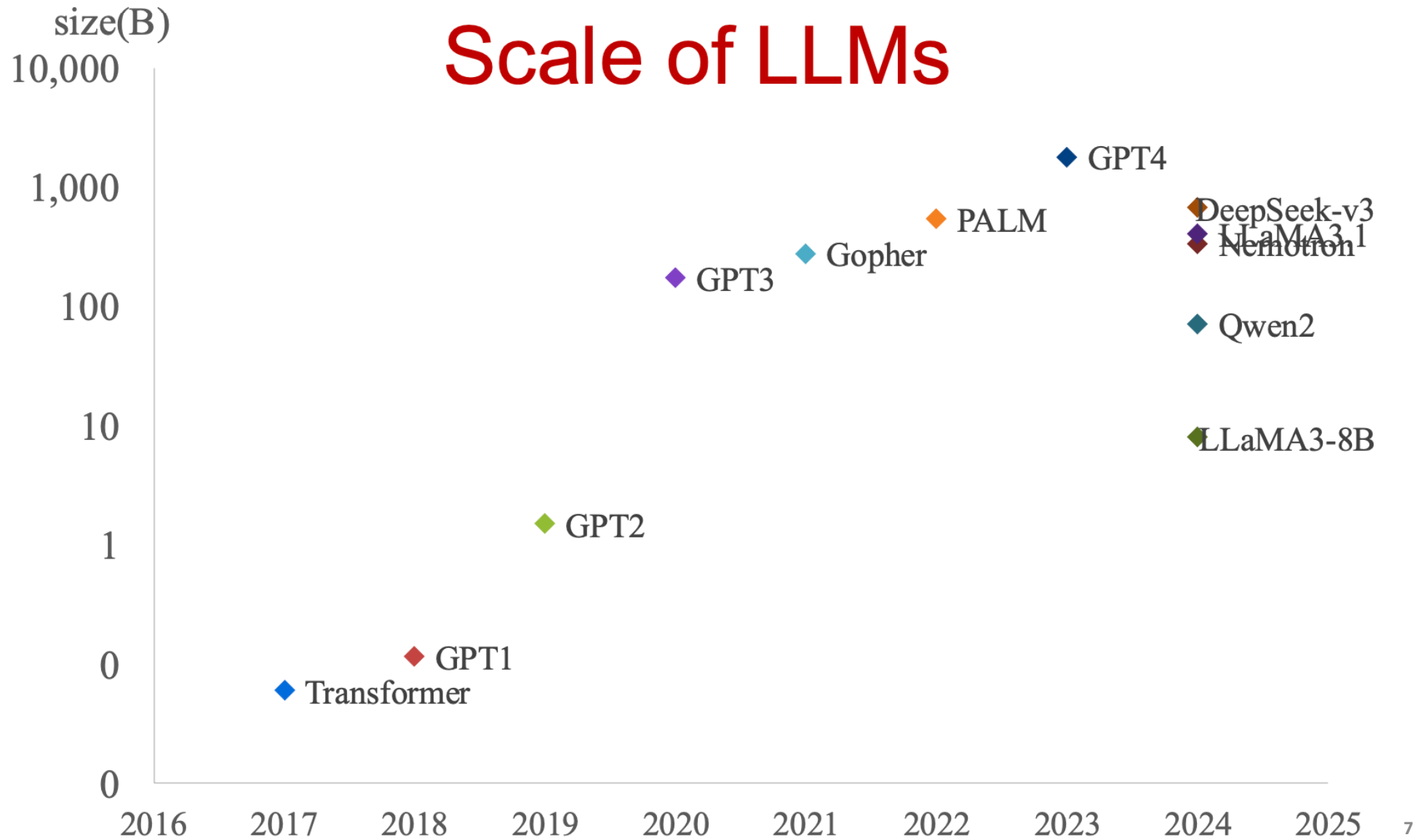
Section 1 – Large Language Models

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History



Scale of LLMs



Capabilities of LLMs



Translation

Google MT



ChatGPT/GPT4



You

Please translate "他上班天天摸鱼"



ChatGPT

The phrase "他上班天天摸鱼" translates to "He slacks off at work every day" in English. Here, "摸鱼" (mō yú) is a Chinese slang term that literally translates to "touching fish," but it's commonly used to mean slacking off or being unproductive, especially at work.

Rewrite Text with Style

 **You**

Please write the following sentence in Yoda's style. Here are a few examples.

You have much to learn. -> Much to learn, you still have.

I would say you have found someone, hmmm? -> Found someone, you have, I would say, hmmm?

You should not drive that fast. ->

 **ChatGPT**

Drive that fast, you should not.

Commonsense Reasoning



You

Fill in the blank: Steven was disgusted that Robert chose to cook pork for dinner, because _ was a vegetarian.

"option1": "Steven", "option2": "Robert"



ChatGPT

The correct answer is "Steven"

Math Reasoning

 **You**

Roger has 4 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

 **ChatGPT**

Roger initially has 4 tennis balls. When he buys 2 more cans of tennis balls, and each can has 3 tennis balls, he adds $2 \text{ cans} * 3 \text{ tennis balls per can} = 6 \text{ tennis balls}$ to his collection.

So, Roger now has a total of $4 \text{ initial tennis balls} + 6 \text{ newly purchased tennis balls} = 10 \text{ tennis balls}$.

Code Generation for a natural language query

 **You**

You are given a 0-indexed array mountain. Your task is to find all the peaks in the mountain array. please write a python program.

 **ChatGPT**

```
def find_peaks(mountain):  
    peaks = []  
    for i in range(1, len(mountain) - 1):  
        if mountain[i] > mountain[i - 1] and  
mountain[i] > mountain[i + 1]:  
            peaks.append(i)  
    return peaks
```



Draw an image about “The power of Large Language Model”



Mathematical Foundations

Secant Lines

Tangent Line

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$
$$f(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h}$$
$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h}$$
$$= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h}$$
$$= \lim_{h \rightarrow 0} h(2x + h)$$
$$= 2x$$

$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$

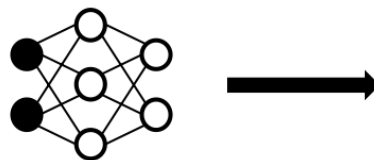
$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$

$f'(a) = \lim_{h \rightarrow 0} \frac{f(a) - f(a-h)}{h}$

Probability Model for Next Token

$$P(\text{next word } y_t \mid \text{Prompt } x, \text{previous words } y_{1:t-1})$$

Santa Barbara has very nice _____



beach	0.5
weather	0.4
snow	0.01

Pittsburgh is a city of _____

bridges	0.6
corn	0.02

Mathematics of Language Model

Probability("Pittsburgh is a city of bridges")
= $P(\text{"Pittsburgh"}) \cdot P(\text{"is"} | \text{"Pittsburgh"})$
· $P(\text{"a"} | \text{"Pittsburgh is"}) \cdot P(\text{"city"} | \dots) \cdot P(\text{"of"} | \dots)$
· $P(\text{"bridges"} | \dots)$

$$\text{Prob.}(x_{1..T}) = \prod_{t=1}^T \frac{P(x_{t+1} | x_{1..t})}{\quad}$$

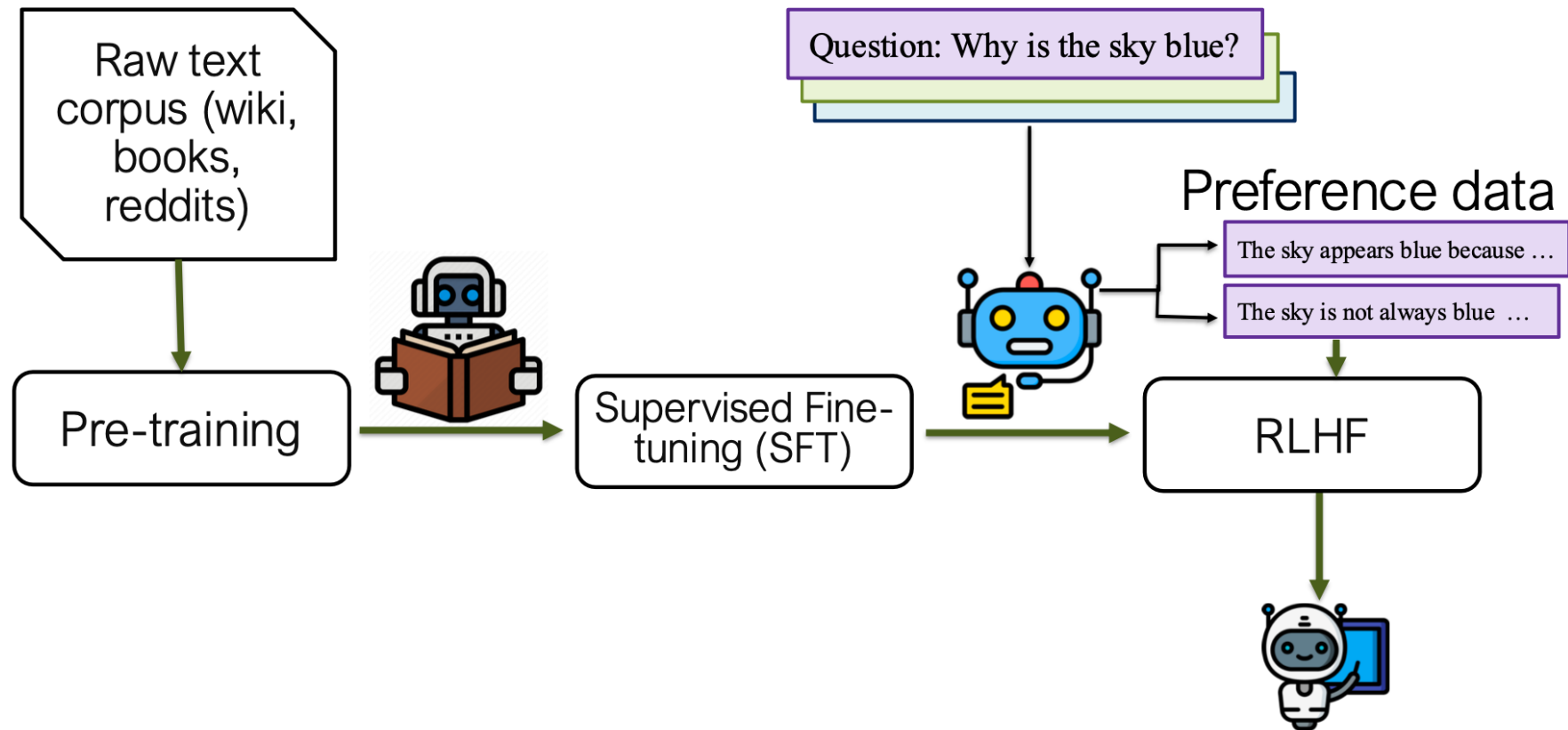

Predicting using Neural Nets
(Transformer network, CNN, RNN)

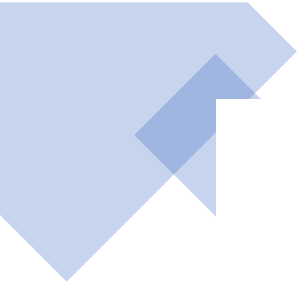
Why is ChatGPT changing AI landscape ?

- **Pre-training** on very large raw data (300B tokens) + small human feedback
- **Instruction following** – easy to use through natural instruction
- **In-context learning** – Generalize well to versatile tasks, by showing a few examples at use time



LLM Learning Framework





Pre-training of Language Models

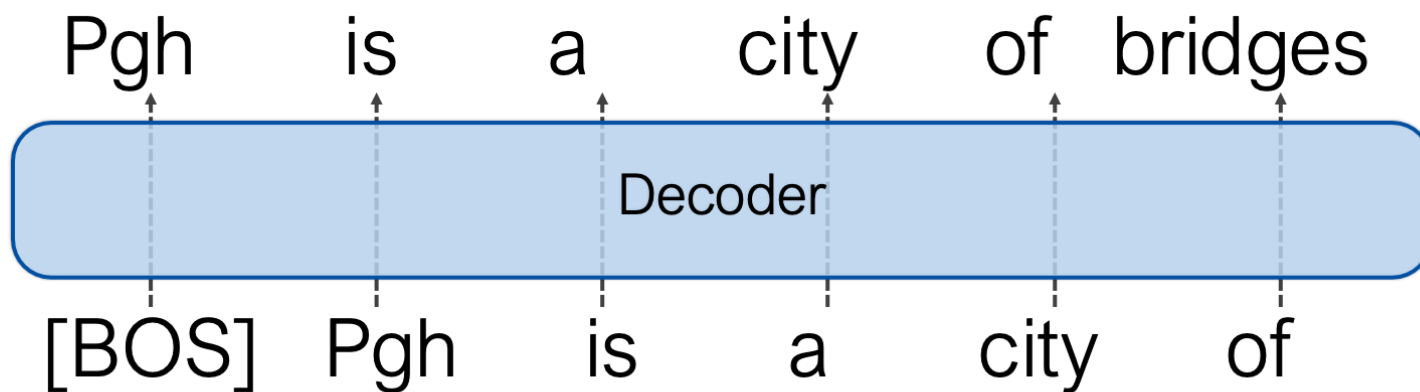


- Data: raw text corpus
 - the larger the better, often crawled from the web
 - quality is important: Wikipedia, books, filtered Reddit posts
- Design of Model Architecture
 - Sparse Attention, Mixture of Expert, etc.
- Training loss:
 - cross-entropy loss for next token prediction.

Training Objective

- Cross entropy loss

$$CE = -\frac{1}{N} \sum_{n=1}^N \log P_{\theta}(x_n | x_{<n})$$



Evaluating performance of Large Language Models

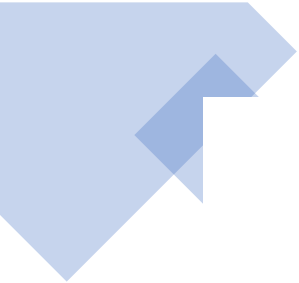
- Perplexity: for open-end generation

$$PPL = \exp - \frac{1}{N} \sum_{n=1}^N \log P_{LM}(x_n | x_{<n})$$

- Reference-based metric:
 - SEScore2/COMET/BLEURT: model-based score
 - BLEU (or ROUGE): based on n-gram matching, old but classic
 - InstructScore: explainable score [Xu et al 2023]

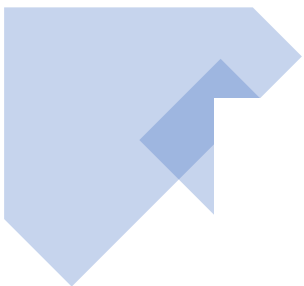
Evaluating performance of Large Language Models

- Using downstream task performance metrics
 - Named entity recognition: F1
 - Question Answering: accuracy, matching rate
 - Code generation: pass-rate @k
 - Retrieval: ndcg
 - Summarization: ROUGE
 - Translation: COMET/SEScore2/BLEU



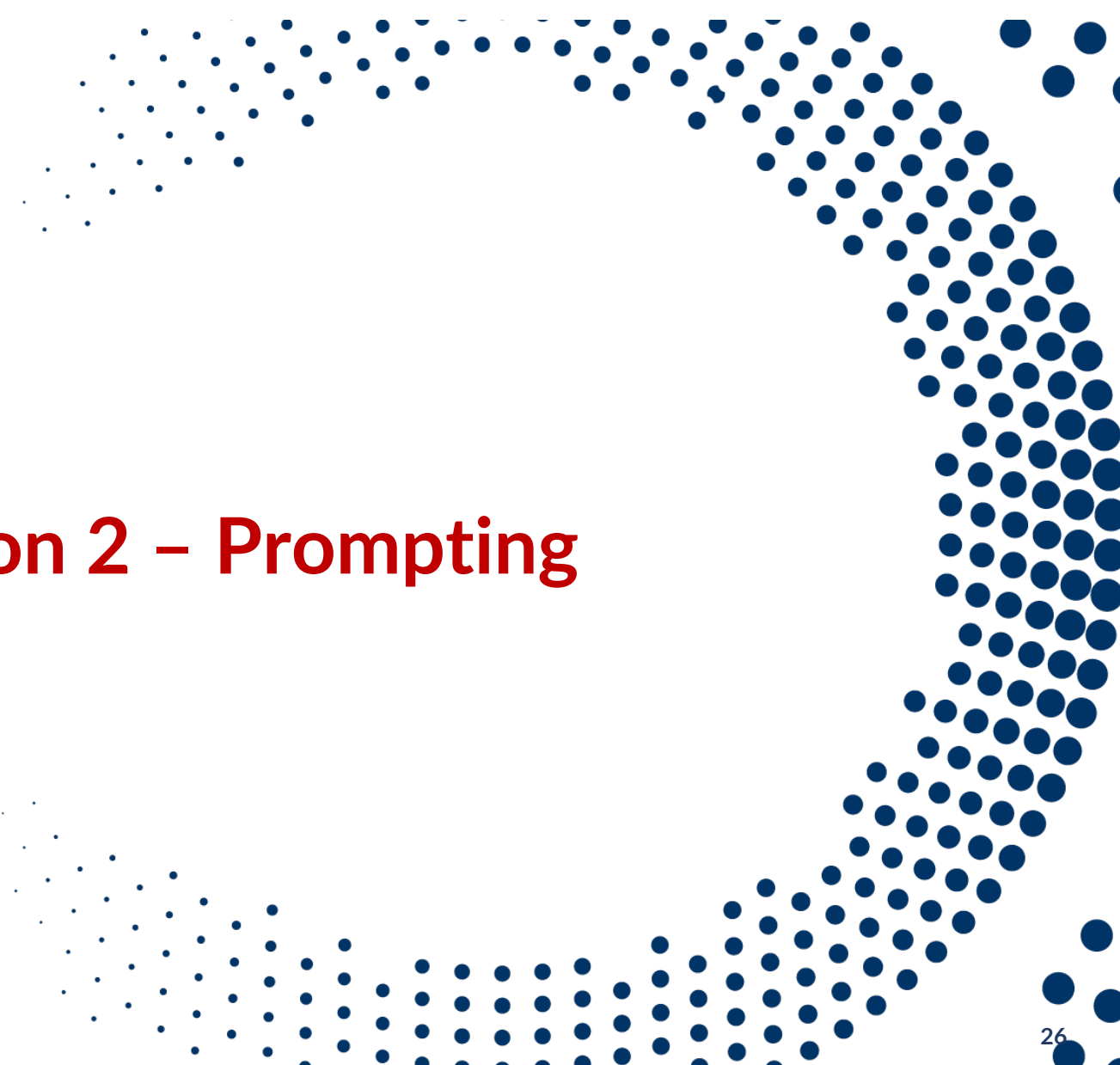
Modern LLM Research

- General
 - Instead of focus on one AI/ML/NLP model, develop generic models that work for many tasks
- Instructibility
 - Interactive system that is able to take natural or infinite commands
- Alignment
 - Not a single explicit objective, but also its behavior aligned with human/society/cultural norm



Programming Models

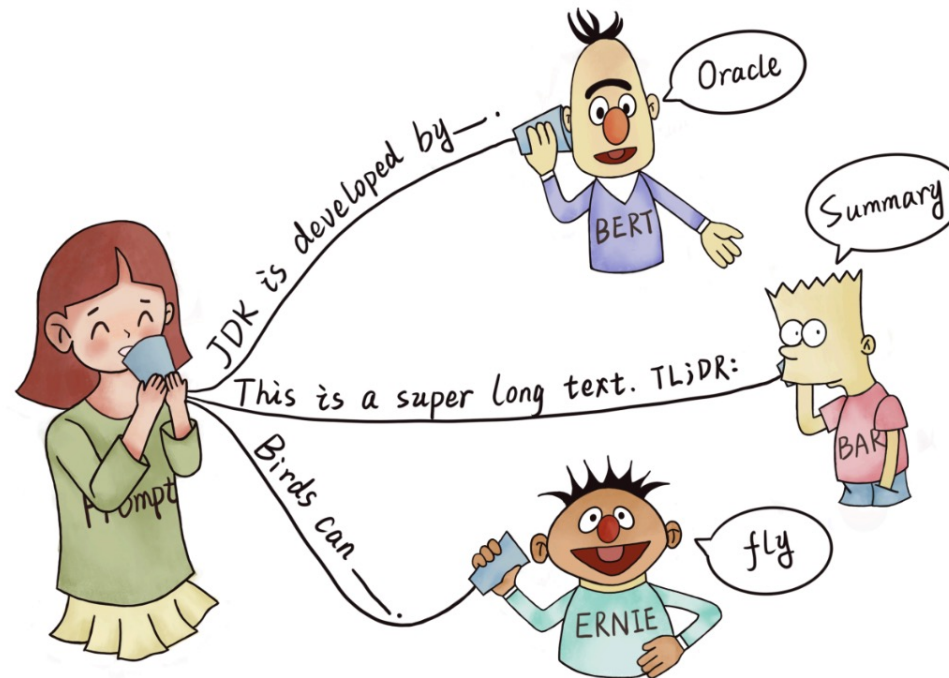
- What are good abstractions to build AI applications?
 - Upper level: integrating models into a product system, tracking and improving product quality over time
 - Middle level: building model training and inference software, streaming dataflow
 - Lower level: building GPU kernels, compilers, etc
- A good abstraction successfully frees the programmer of one or more concerns (e.g., performance or failures) while supporting a wide range of apps on top



Section 2 – Prompting

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What is prompting?



Encouraging a pre-trained model to make particular predictions by providing a textual “prompt” specifying the task to be done.

Basic Prompting

x = **When a dog sees a squirrel, it will usually**

(GPT-2 Small) be afraid of anything unusual. As an exception, that's when a squirrel is usually afraid to bite.

(GPT-2 XL) lick the squirrel. It will also touch its nose to the squirrel on the tail and nose if it can.

Append a textual string to the beginning of the sequence and complete



Standard Prompting Workflow

- Fill a prompt template
- Predict the answer
- Post-process the answer



Prompt Template

Input: $x = \text{"I love this movie"}$



Template: $[x]$ Overall, it was $[z]$



Prompting: $x' = \text{"I love this movie. Overall it was [z]"}$

A template where you fill in with an actual input



Chat Prompts

- Recently, many models are trained as chatbots
- Usually inputs are specified in OpenAI messages format

```
messages=[  
  {  
    "role": "system",  
    "content": "Please classify movie reviews as 'positive' or 'negative'."  
  },  
  {  
    "role": "user",  
    "content": "This movie is a banger."  
  },  
]
```

- Roles:
 - **“system”**: message provided to the system to influence behavior
 - **“user”**: message input by the user
 - **“assistant”**: message output by the system



Chat Prompts Behind the Scenes

- Behind the scenes, messages are converted to token strings

LLaMa

```
Sys. [INST]
      <<SYS>>
      You are an assistant that ...
      <</SYS>>
      [/INST]

User| [INST]This movie is great.[/INST]

Asst.| Positive.
```

Alpaca

```
Sys. | ### Instruction:
      You are an assistant that ...

User| ### Instruction:
      This movie is great.

Asst.| ### Response:
      Positive.
```

Answer Prediction

- Given a prompt, predict the answer

Prompting: $x' =$ "I love this movie. Overall it was [z]"



Predicting: $x' =$ "I love this movie. Overall it was fantastic"

- Use any inference algorithms, as in generation class



Post-processing

- Based on the answer, select the actual output
- For instance:
 - Taking the output as-is
 - Formatting the output for easy visualization
 - Selecting only parts of the output that you want to use
 - Mapping the outputs to other actions



Output Formatting

- For user-facing applications, format in a pretty way

Markdown Rendering

GN You

Please write a table with the last 5 presidents and their birthdates.

ChatGPT

Here is a table with the last 5 U.S. Presidents and their birthdates:

President	Birthdate
Joe Biden	November 20, 1942
Donald Trump	June 14, 1946
Barack Obama	August 4, 1961
George W. Bush	July 6, 1946
Bill Clinton	August 19, 1946

Code

GN You

Output that in JSON format.

ChatGPT

Here is the information in JSON format:

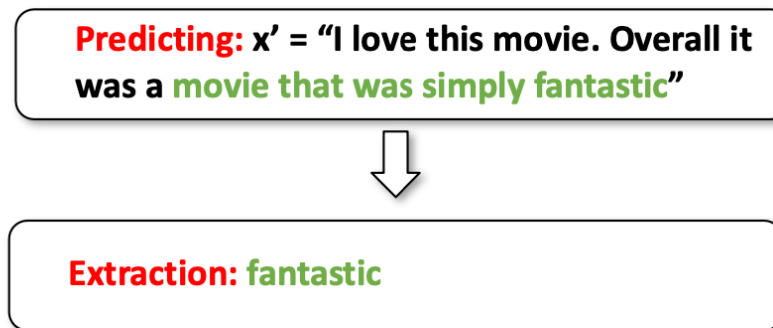
```
json Copy code

[
  {"President": "Joe Biden", "Birthdate": "November 20, 1942"},
  {"President": "Donald Trump", "Birthdate": "June 14, 1946"},
  {"President": "Barack Obama", "Birthdate": "August 4, 1961"},
  {"President": "George W. Bush", "Birthdate": "July 6, 1946"},
  {"President": "Bill Clinton", "Birthdate": "August 19, 1946"}
]
''' &#8203;''' [oacite:0] '''&#8203;'''
```



Output Selection

- From a longer response, select the information indicative of an answer



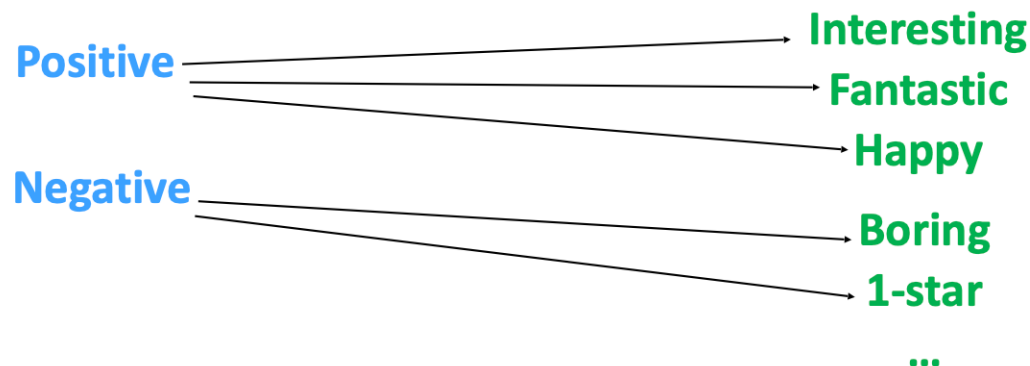
- Various methods for extraction
 - **Classification:** identify keywords
 - **Regression/numerical problems:** identify numbers
 - **Code:** pull out code snippets in triple-backticks

Output Mapping

- Given an answer, map it into a class label or continuous value



- Often map many extracted words onto a single class





Prompting Techniques

Few-shot Prompting / In-context Learning

Instruction | Please classify movie reviews as 'positive' or 'negative'.

Examples

Input: I really don't like this movie.
Output: negative

Input: This movie is great!
Output: positive

Provide a few examples of the task together with the instruction



LLMs are sensitive to Small Changes in In-Context Examples

- Example ordering (Lu et al. 2021)
- Label balance (Zhang et al. 2022)

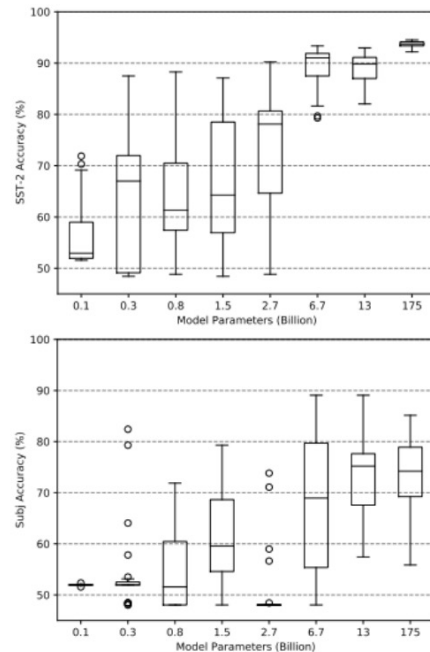
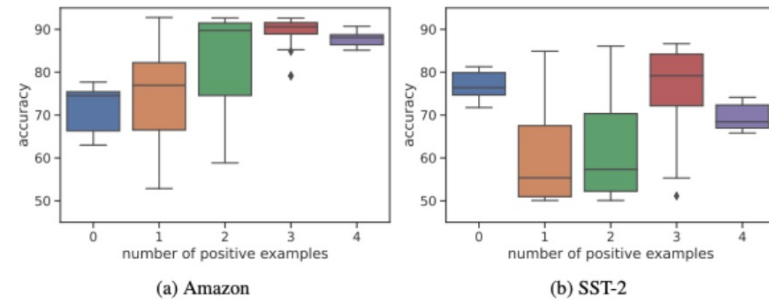
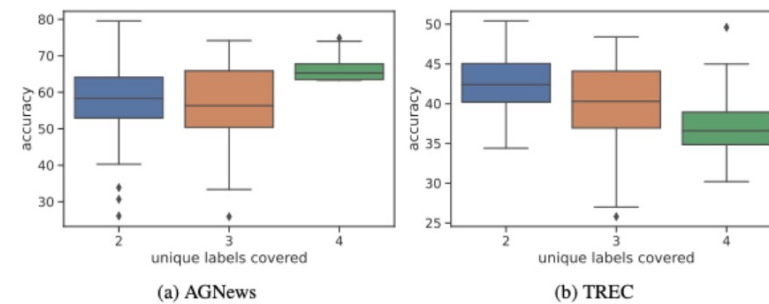


Figure 1: Four-shot performance for 24 different sample orders across different sizes of GPT-family models (GPT-2 and GPT-3) for the SST-2 and Subj datasets.



- Label coverage (Zhang et al. 2022)



Chain of thought Prompting

- Get the model to explain its reasoning before making an answer

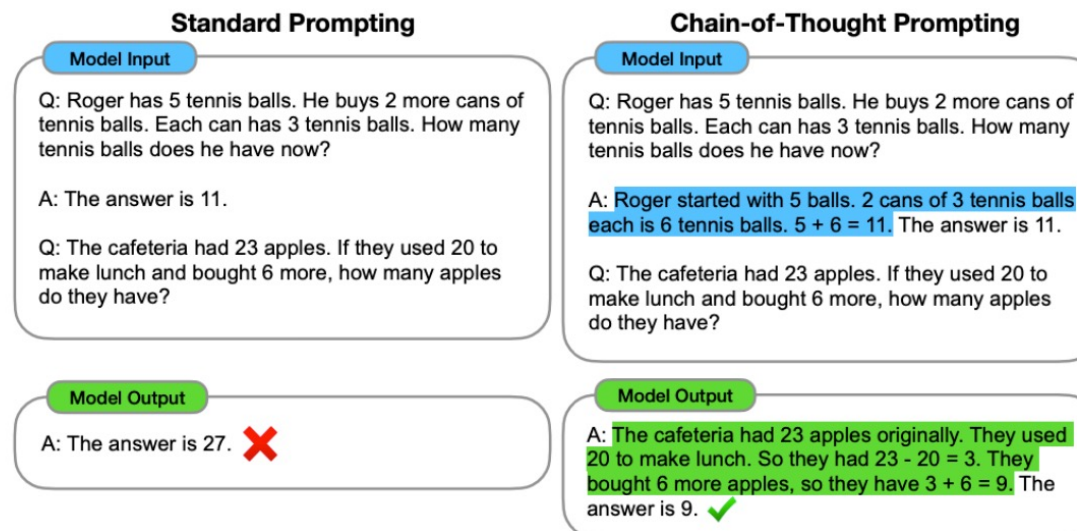


Figure 1: Chain-of-thought prompting enables large language models to tackle complex arithmetic, commonsense, and symbolic reasoning tasks. Chain-of-thought reasoning processes are highlighted.

- Provides the model with adaptive computation time

Unsupervised Chain-of-thought Prompting

- Just adding a prompt that encourages the model to explain decisions can induce reasoning

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The answer is 8. ✗

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. ✓

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 ✗

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

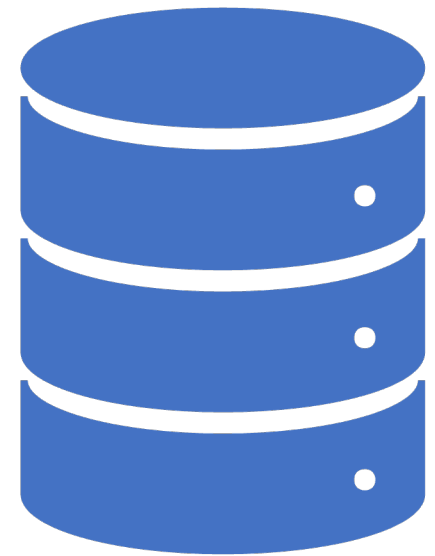
A: **Let's think step by step.**

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

- Note: GPT models reason even w/o specific instructions now (probably due to instruction tuning)



Retrieval Augmented Generation



Problems

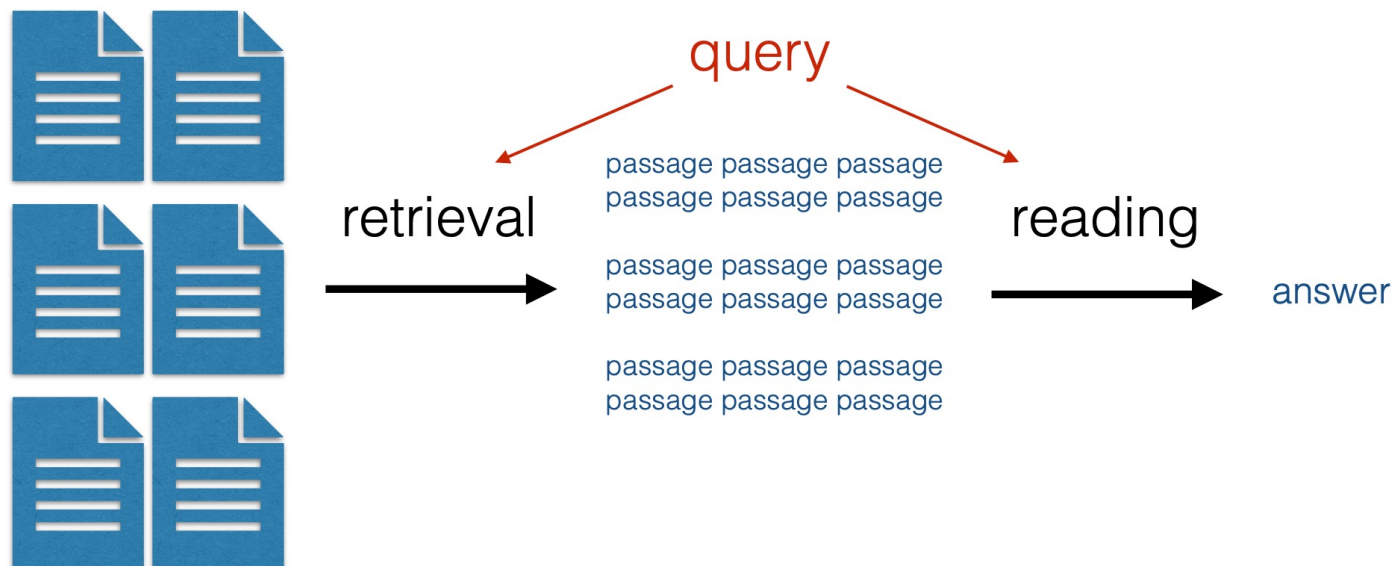
- Accuracy issues:
 - Knowledge cutoffs: parameters are usually only updated to a particular time
 - Private data: data stored in private text or data repositories not suitable for training
 - Learning features: even for data that the model was trained on, it might not be sufficient to get the right answer
- Verifiability issues:
 - It is hard to tell if the answer is correct or not



Retrieval-augmented Generation

(Chen et al. 2017)

- **Retrieve** relevant passages efficiently
- **Read** the passages to answer the query





Retrieval Methods



- **Sparse retrieval**
- **Document-level dense retrieval**
- **Token-level dense retrieval**
- **Cross-encoder reranking**
- **Black-box retrieval** (just ask Google/Bing)

Sparse Retrieval

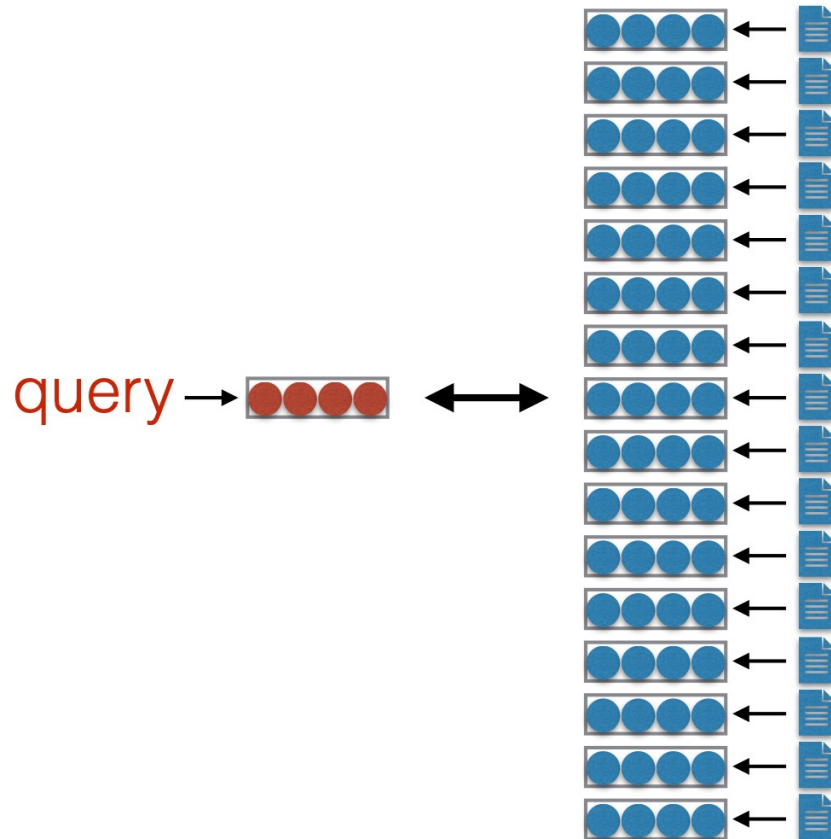
- Express the query and document as a sparse word frequency vector (usually normalized by length)

q=what is nlp		$d_1 =$ what is life ? candy is life !	$d_2 =$ nlp is an acronym for natural language processing	$d_3 =$ I like to do good research on nlp
what	0.33	0.25	0	0
candy	0	0.125	0	0
nlp	0.33	0	0.125	0.125
is	0.33	0.25	0.125	0
language	0	0	0	0
...	...	...	...	...
		$q \cdot d_1 = 0.165$	$q \cdot d_2 = 0.0825$	$q \cdot d_3 = 0.0413$

- Find the document with the highest inner-product or cosine similarity in the document collection

Dense Retrieval

- Encode document/query and find nearest neighbor
- Can use:
 - Out-of-the-box embeddings
 - Learned embeddings



Learning Retrieval-oriented Embeddings

- Select positive and negative documents, train using a contrastive loss (e.g. hinge loss)

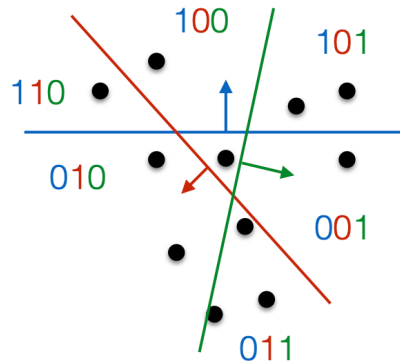
$$\mathcal{L}(\theta, q) = \sum_{d_{\text{pos}} \in D_{\text{pos}}} \sum_{d_{\text{neg}} \in D_{\text{neg}}} \max(0, s(q, d_{\text{neg}}; \theta) - s(q, d_{\text{pos}}; \theta))$$

- **DPR** (Karpukhin et al. 2020): learn encoders based on a BM25 hard negatives and in-batch negatives.
- **Contriever** (Izacard et al. 2022): contrastive learning using two random spans as positive pairs

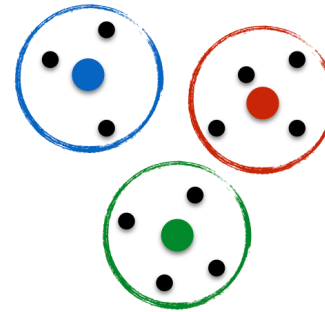
Approximate Nearest Neighbor Search

- Methods to retrieve embeddings in sub-linear time

Locality sensitive hashing:
make partitions in continuous space, use like inverted index



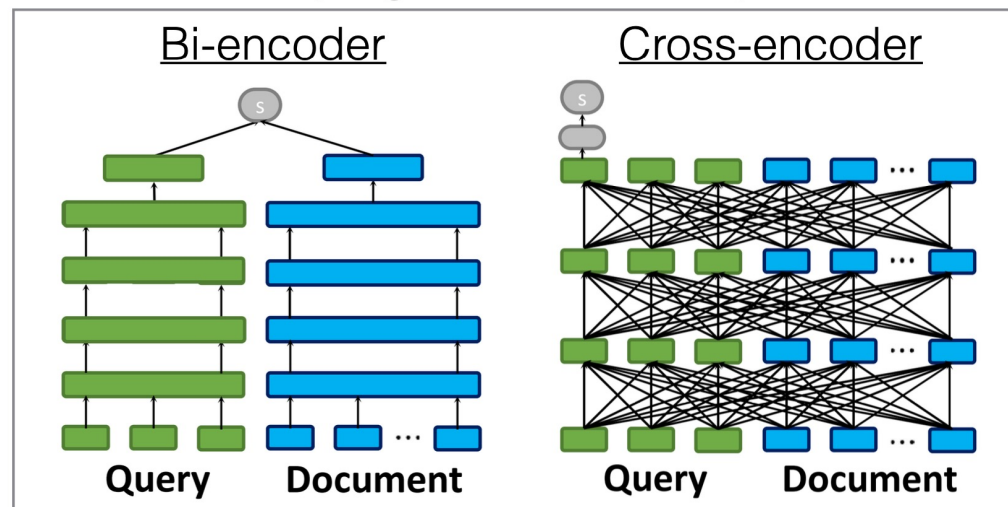
Graph-based search: create
“hubs” and search from there



- Software: FAISS, ChromaDB

Cross Encoder Re-ranking

- Jointly encode both queries and documents using neural model (Nogueira et al. 2019)

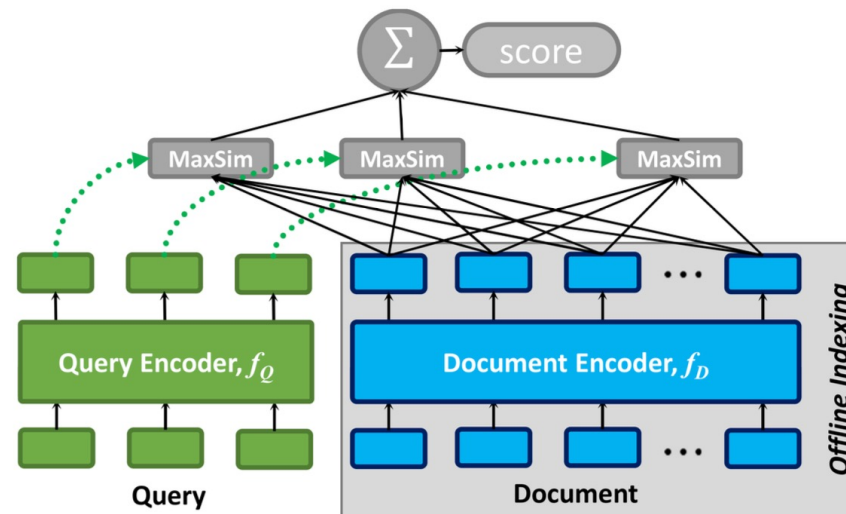


- Precludes approximate nearest neighbor lookup, so can only be used on small number of candidates

Figure from Khattab et al. (2020)

Token-level Dense Retrieval

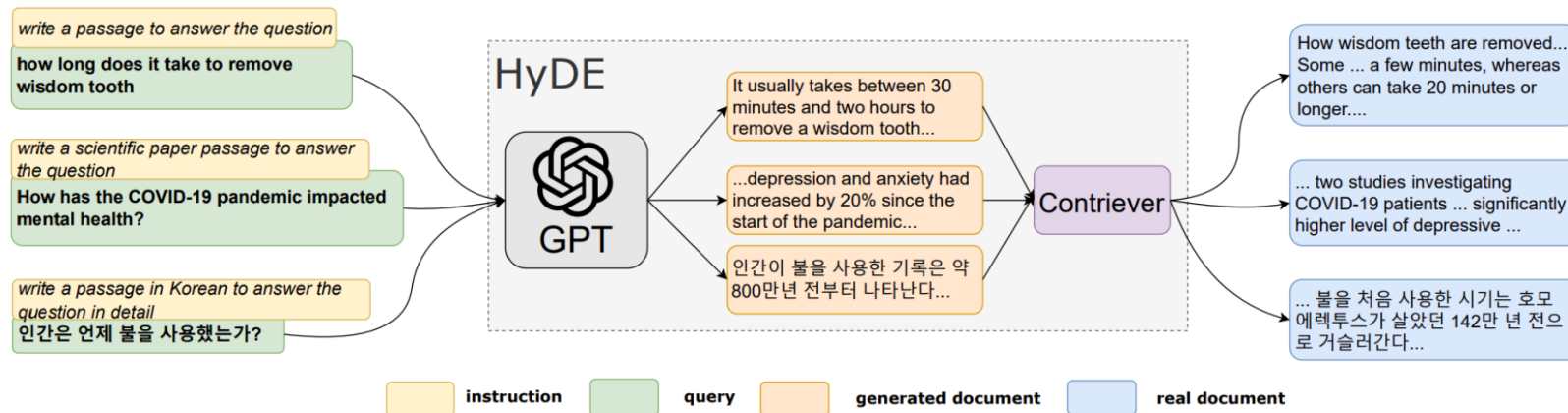
- ColBERT (Khattab et al. 2020) use contextual representations of all query and document tokens to compute retrieval score.

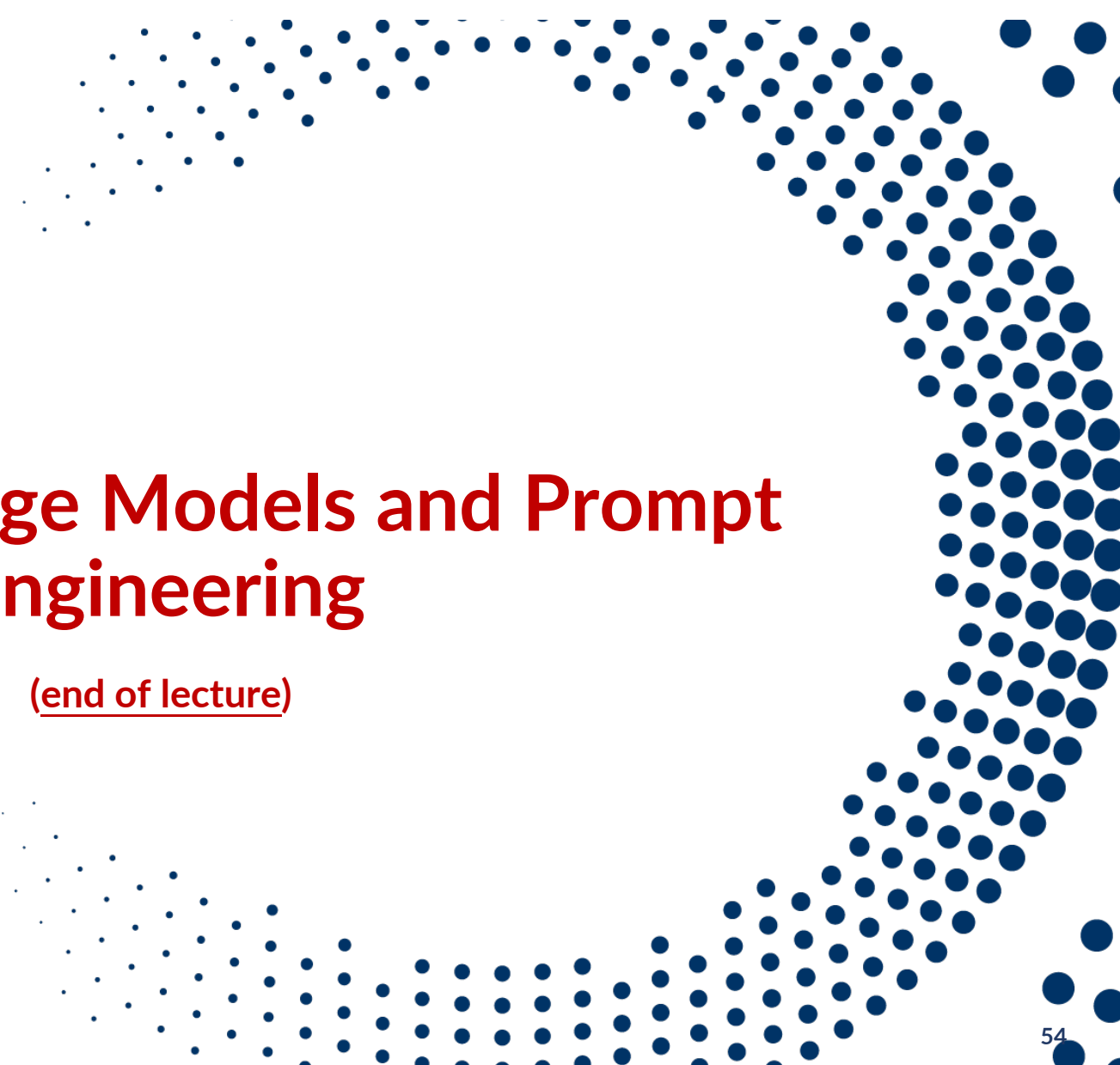


- Significantly more effective (but more costly) than single-vector retrieval

Hypothetical Document Embeddings

- Generate a “hypothetical document” for the query using an LLM, and try to look it up
- Can be easier than trying to match under-specified query





Large Language Models and Prompt Engineering

(end of lecture)

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